Basic Counting - Teacher Diagnostic Key

Integrated Grade-9 learner pack

COMB-01 - Teacher Diagnostic Key

Teacher/control artifact. Internal topic and support-control language is allowed here; it is not included in the student PDF.

Governing diagnostic

A wrong numerical count should be classified before correcting arithmetic:

1. counted-object/identity error;
2. ordered-vs-unordered error;
3. non-disjoint or non-exhaustive case split;
4. stage/product error;
5. repeated-object overcount;
6. restriction applied too late;
7. complement universe mismatch;
8. inclusion-exclusion overlap error;
9. counting/arithmetic ownership collision.

Historical anchors

- IOQM-2025-Q05 = 45: sum admissible pair counts 9+8+...+1.
- IOQM-2025-Q15 = 40: restricted injection of three coupon-pairs into distinct envelopes; allowed sets {3,4,5,6}, {1,2,5,6}, {1,2,3,4}. IE check: 120-120+48-8=40.
- IOQM-2025-Q18 = 40: C(9,2)*C(6,2)=540, requested remainder mod 100 is 40.
- IOQM-2024-Q02 = 12: 2*3!.
- IOQM-2023-Q07 = 48: fix the 1/2 axis, six cyclic orders, 2^3 opposite-pair colour choices.
- IOQM-2023-Q17 = 66: fourth order statistic expectation 200/3, requested floor 66.
- IOQM-2023-Q20 = 43: factor cardinality/max constraints, then binomial counts; independent verification gives N=439, requested 4+39.

Practice answers

1. 12 — product of 3 shirt and 4 cap choices.
2. 8 — disjoint alternatives 5+3.
3. 15 — C(6,2).
4. 24 — 4!.
5. 12 — 4!/2!.
6. 12 — choose units digit 2 or 4, then 3!.
7. 56 — C(8,3).
8. 60 — 6!/(3!2!).
9. 60 — 5*4*3.
10. 10 — choose positions of two ones: C(5,2).
11. 300 — first digit 5 choices, then 5*4*3.
12. 360 — symmetry: half of 6! have 1 before 2.
13. 128 — choose which of 1,2 is present, then any subset of remaining six: 2*2^6.
14. 60 — ordered injection 5*4*3.
15. 61 — 5^3-4^3.

16. 1260 — 7!/(2!2!).
17. 1260 — last digit 0 contributes 6*5*4*3=360; last digit 2/4/6 contributes 3*(5*5*4*3)=900.
18. 140 — C(10,4)-C(8,4)=210-70.
19. 3600 — 7!-2*6!.
20. 30 — the two pairwise order constraints divide 5! by 4.
21. 9 — derangements of four; IE gives 24-24+12-4+1=9.
22. 60 — 50+20-10.
23. 1560 — 4^6-4*3^6+6*2^6-4.
24. 30 — fix one 3 last, arrange 11223: 5!/(2!2!).
25. 120 — total 7!/(3!2!2!)=210; bad first-1 arrangements 6!/(2!2!2!)=90.
26. 456 — C(4,2)C(8,3)+C(4,3)C(8,2)+C(4,4)C(8,1).
27. 25 at least one; 5 neither — 18+16-9=25.
28. 360 — choose 2 of the 3 even and all 3 odd digits, then arrange: C(3,2)*5!.
29. 60 — total 6!/(2!2!2!)=90; bad AA-block 5!/(2!2!)=30.
30. 540 — 3^6-3*2^6+3.

Independent mastery answers

1. 56 — ordered roles 8*7.
2. 28 — unordered C(8,2).
3. 12 — last digit 1 or 3, then 3!.
4. 30 — 5!/(2!2!).
5. 127 — 2^7-1.
6. 60 — 40+30-10.
7. 105 — C(9,4)-C(7,2)=126-21.
8. 600 — first digit 5 choices, then 5*4*3*2.
9. 9 — derangements of four.
10. 30 — fix one 2 last, arrange 11223.
11. 175 — C(10,4)-C(7,4)=210-35.
12. 90 — each of three independent pair-order constraints halves 6!: 720/8.
13. 150 — 3^5-3*2^5+3.
14. 60 — 6!/(3!2!).
15. 360 — C(3,2)*C(3,3)*5!.
16. 36 — C(8,3)-C(6,3)=56-20.

H-control fading map

H3: counted object, representation and first computation line supplied.

H2: counted object/representation supplied; computation withheld.

H1: only the distinguishing clue is supplied.

H0: no route label or default hint.

These H-levels are teacher controls and must not appear in the learner export.